

PEOPLE'S DEMOCRATIC REPUBLIC OF ALGERIA
MINISTRY OF HIGHER EDUCATION AND SCIENTIFIC RESEARCH
THE NATIONAL SCHOOL OF ARTIFICIAL INTELLIGENCE

Advancing Legal Reasoning in Algerian Law

*Integrating RAG, Knowledge Graphs & Argument Mining
for Citation-Faithfulness*

Ibrahim El Khalil **Attia** · Maab **Chaoui**

SUPERVISION

Dr. Fouad Dahak (ENSIA) · Prof. Iván Cantador (U. Autónoma de Madrid)

JURY

Dr. Wahid Chami (ENSIA), President · Dr. Mohamed Brahimi (ENSIA), Examiner

PUBLIC DEFENSE · 13 JUNE 2026 · FYP/2025/135

Mata

V.

Avianca, Inc.

S.D.N.Y. · 2023

A personal-injury suit against an airline. The plaintiff’s lawyer filed a brief drafted with **ChatGPT**, citing six judicial decisions in support — and when the court grew suspicious, the model **vouched that they were real.**

THE HALLUCINATED
CASES

~~Varghese v. China Southern Airlines~~

~~Shaboon v. EgyptAir~~

~~Petersen v. Iran Air~~

~~Martinez v. Delta Air Lines~~

~~Estate of Durden v. KLM~~

~~Miller v. United Airlines~~

THIS DEFENSE

Outline

I. Landscape & Limits

why legal AI fails — and why Algeria is the hard case

II. Foundations

four pillars of citation-faithful reasoning

III. AKN-RLM

the solution, layer by layer

IV. Evaluation & Results

AlgerianLegalBench, baselines, ablations

V. Conclusion

what we claim — and what we do not

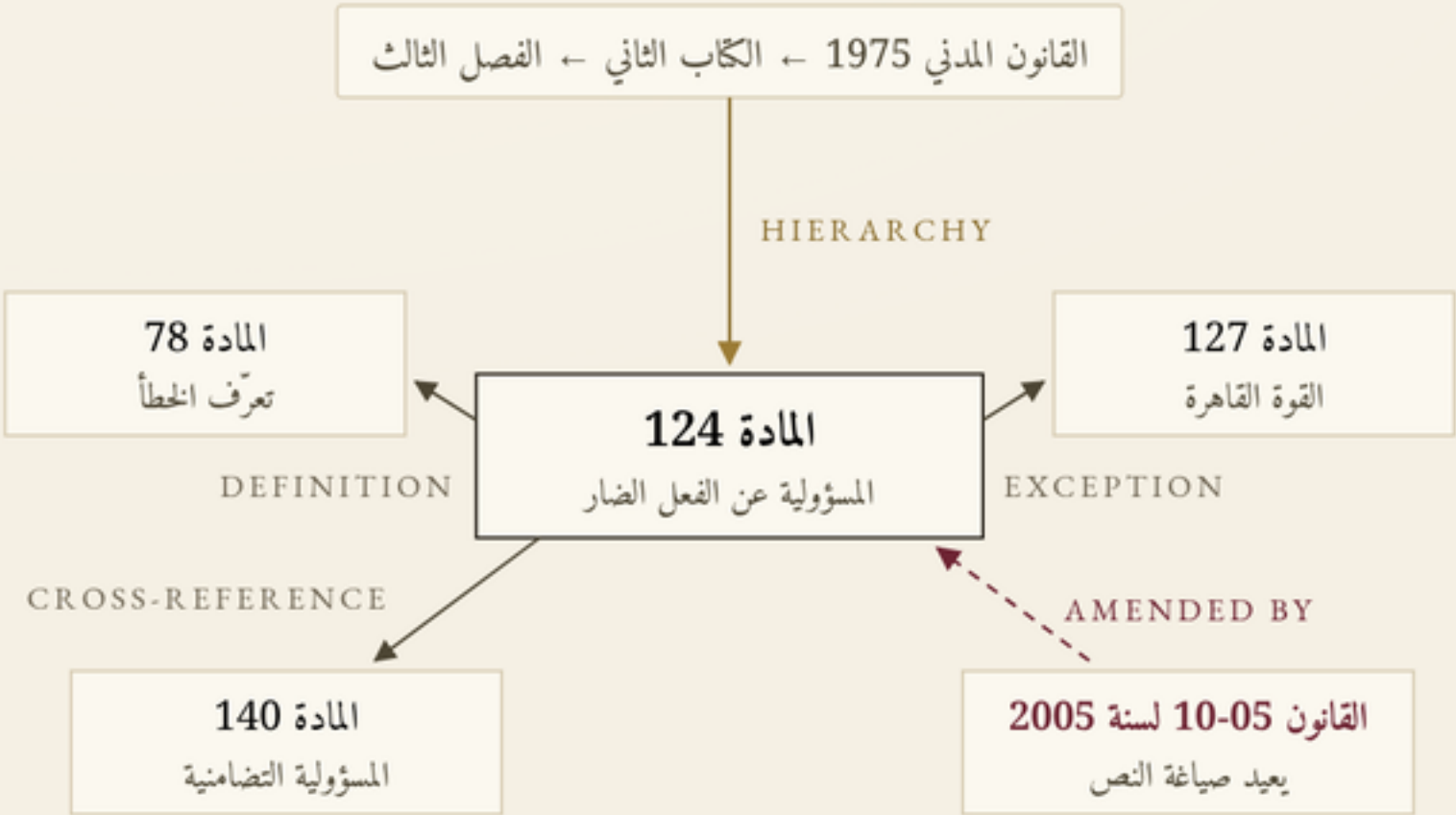
Landscape & Limits

why legal AI fails — and why Algeria is the hard case

THE LEGAL DOMAIN THROUGH AN AI LENS

Legal meaning is not *linguistic* meaning

- i* **Structured** — codes, chapters, numbered articles; structure carries meaning
- ii* **Interdependent** — definitions, exceptions and cross-references live in other articles
- iii* **Time-sensitive** — amendments decide which version is law *today*
- iv* **Authority-bound** — an answer must cite the provision that makes it true



a provision means nothing without the graph around it

HENCE *Structure is an asset when machine-readable — and a liability when buried inside publication-oriented PDFs.*

THE PROBLEM

In law, fluency is not the standard. *Justification* is.

THE STANDARD OF A LEGAL ANSWER

a real legal source · the correct article · a valid citation — *or a principled abstention*

FAILURES OF LEGAL AI

FAILURE I — HALLUCINATION

Inventing the law

Models fabricate claims, citations and whole authorities while sounding confident — *Mata v. Avianca* shows fabricated citations entering real legal submissions.

Ji et al. 2023 · Mata 2023

FAILURE II — THE ATTRIBUTION GAP

Answering without grounding

Even a *true* answer is legally weightless if the system cannot show **which source supports which claim**. Attribution is a requirement, not a feature.

Rashkin et al. 2023

FAILURE III — THE HALF-MEASURE

Citing without justifying

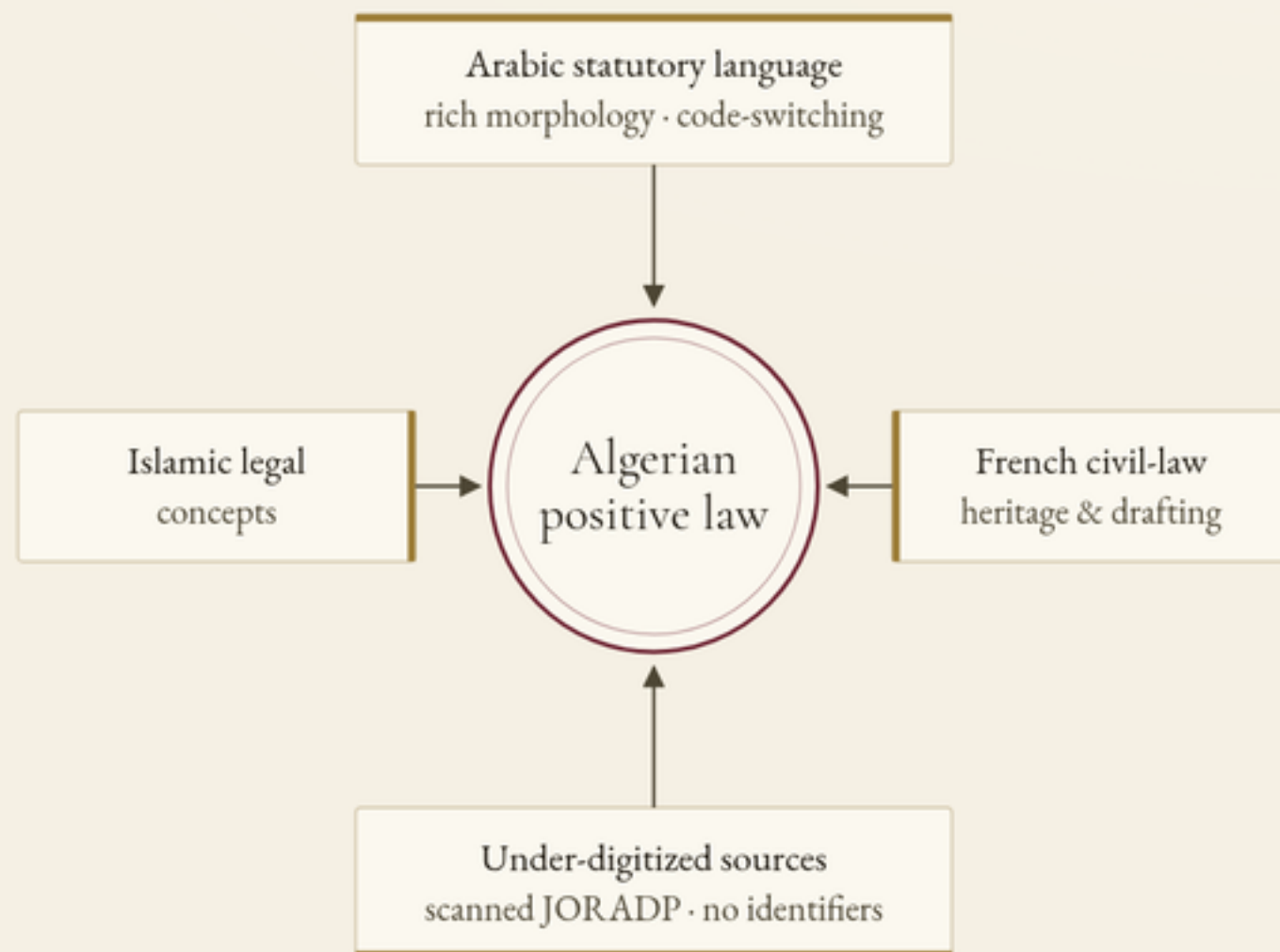
Citation-aware generation is improving — but an emitted citation does not guarantee the article **exists**, is **in force**, or actually **supports** the claim.

Zhang et al. 2025

Fluency, truth, attribution, justification — each is a strictly stronger requirement. Law demands the last one.

THE LOW-RESOURCE CONDITION

Algerian law as a *stress test*



- i* **English legal AI is well-resourced** — benchmarks, corpora, encoders: LexGLUE · LegalBench · COLIEE
- ii* **Arabic is not — Algeria least of all** — the closest encoder, AraLegal-BERT, has $\approx 3.6\%$ Algerian coverage
- iii* **Infrastructure must come first** — a structured corpus and a benchmark, before any system can be deployed

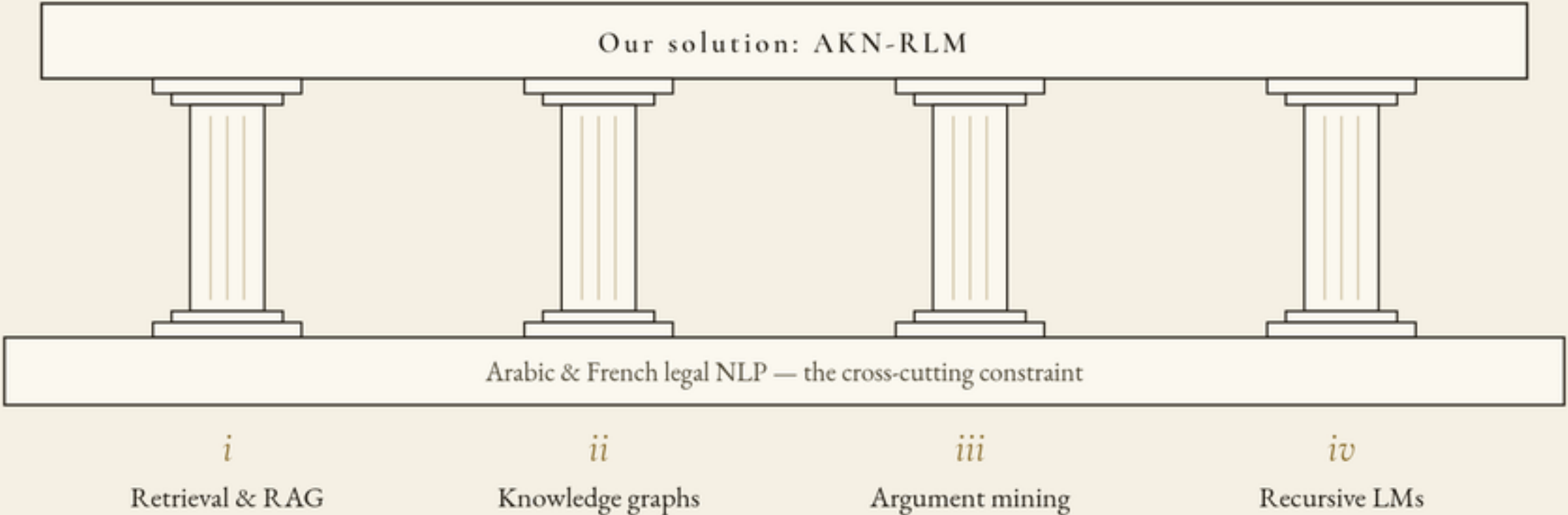
THE RESEARCH QUESTION

Can we build a legal AI system for Algerian law that remains *citation-faithful*, *jurisdiction-aware*, and capable of *principled abstention*?

EACH PROPERTY IS MEASURABLE — CITATION F1 · HALLUCINATED CITATIONS · JURISDICTIONAL INFLECTION · ABSTENTION F1

Foundations

the pillars our work stands on



THE NON-ENGLISH GAP

Arabic legal NLP — a *cross-cutting* constraint

MODEL	NATURE	ROLE IT CAN PLAY HERE
AraBERT · CAMeLBERT	Arabic encoders	Arabic understanding, classification, retrieval encoders
CamemBERT	French encoder	the French half of Algerian legal text
XLM-R	cross-lingual encoder	bridges the bilingual corpus
Jais	Arabic generative LLM (70B)	Arabic-first generation
AraLegal-BERT	legal Arabic encoder	closest fit — yet its Algerian coverage remains limited

No single model covers Arabic, French, legal register and generation — the language problem must be solved by the architecture.

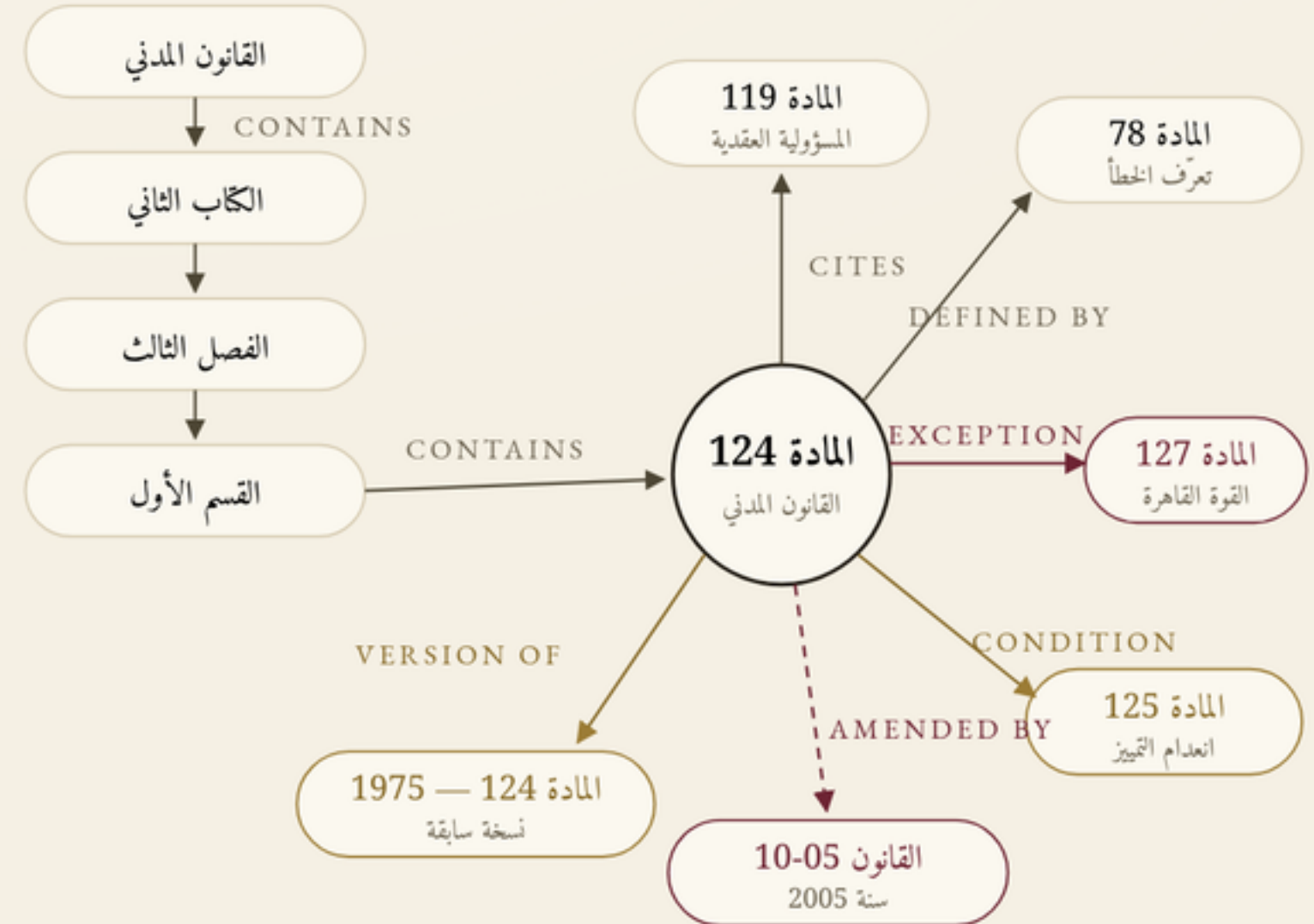
Retrieval & RAG — grounding, not yet *reasoning*

BM25 Lexical exact legal identifiers: article numbers, law numbers, dates, institutions. <i>Robertson & Zaragoza 2009</i>	DENSE Semantic lay phrasing, paraphrase, mixed Arabic/French expressions. <i>Karpukhin et al. 2020</i>	HYBRID + RAG Grounded fuse channels that fail differently; generate <i>from retrieved law</i>, not from model memory. <i>Lewis et al. 2021</i>	SELF-CORRECTIVE Verified Self-RAG, Corrective RAG, Speculative RAG: grounded generation needs verification loops. <i>Asai 2023 · Yan 2024 · Wang 2025</i>
--	---	---	---

RAG retrieves relevant passages — it does not guarantee article correctness, completeness, temporal validity, or claim-level support.

Legal Knowledge Graphs — because law is *relational*

- Provisions **cite**, **amend**, **define**, **limit** and **depend on** other provisions — a graph, not a list of passages.
- GraphRAG**: relations guide evidence expansion where text similarity is blind — exceptions and amendments never “resemble” the query. Edge et al. 2025
- Temporality** is the open challenge: the graph must answer “what was the law *on this date*?” Kucuk & Can 2025
- Ours is **derived from the structured corpus** — not manually invented.



PILLAR II *Typed edges — hierarchy · cites · defines · conditions · excepts · amends · version — supply what similarity cannot.*

PILLAR III OF IV · INSPECTABILITY

Argument-aware evidence — not flat text, *structured units*

- Retrieved provisions become **ADU-inspired triplets** — { source · relation · statement } — instead of undifferentiated context.
- Legal roles survive generation: **rule, condition, exception** — a fluent answer can no longer collapse them.
- The same units later feed **Gate III**: claim-level support is checked against these grounds by NLI.

SCOPE — DELIBERATELY BOUNDED

Not a full argument-mining system. No annotated Algerian argument corpus exists; this is a lightweight structuring layer, validated indirectly through the registry and the faithfulness gates.

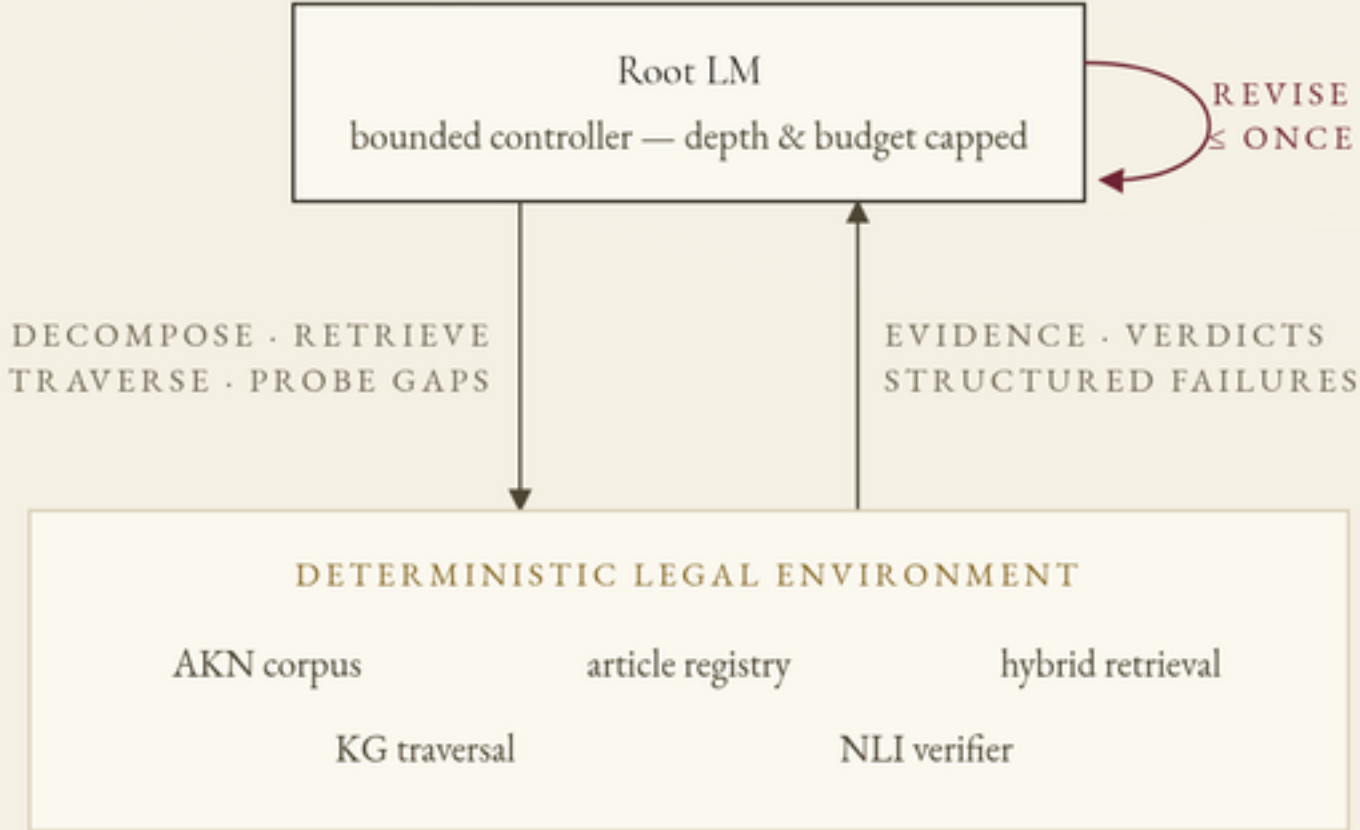
EVIDENCE STATE FOR — “must the driver compensate the victim?”

Art. 124 Civil Code	SUPPORTS →	“fault that causes damage to another obliges reparation”	MAIN RULE
Art. 124 Civil Code	CONDITIONS →	“liability attaches only where fault, damage and causation are all present”	CONDITION
Art. 127 Civil Code	LIMITS →	“no liability where the damage arises from force majeure”	EXCEPTION

after Toulmin 1958 · Walton 2008 · ASPIC* 2014 — borrowing the shared insight, not the full machinery.

PILLAR IV OF IV · CONTROL

Recursive Language Models — the *orchestration* layer



the model queries the law — it never becomes the law

- The RLM can **decompose** the question, **inspect** evidence, **call** retrieval, **traverse** the graph, **verify** support, and **revise** its path. Zhang et al. 2025
- A **control abstraction** over retrieval, RAG and the KG — not a replacement for them.
- **Not an autonomous legal judge:** every step is bounded, logged, and grounded in deterministic infrastructure.

PILLAR IV *Recursion buys the right to look again — under a budget, over infrastructure that cannot hallucinate.*

SYNTHESIS — GAPS & CONTRIBUTIONS

Every ingredient exists. *No system unites them.*

1

Structured Algerian corpus

Arabic legal NLP exists — but no machine-readable Algerian articles with stable identifiers

→ Akoma Ntoso corpus, 171 files (1963–2025)

2

Temporal legal KG

KGs give relational structure — none covers Algerian references, amendments, versions

→ RDF Legal Knowledge Graph

3

Algerian benchmark

no citation-grounded evaluation exists for Algerian law

→ AlgerianLegalBench — 244 questions

4

Recursive orchestration

RAG grounds generation — but single-pass pipelines cannot inspect or revise

→ bounded RLM controller

5

Argument-aware evidence

argument mining is inspectable — but never integrated into grounded verification

→ Toulmin ADU structuring

6

Unified faithful architecture

each piece is mature in isolation — no citation-faithful whole

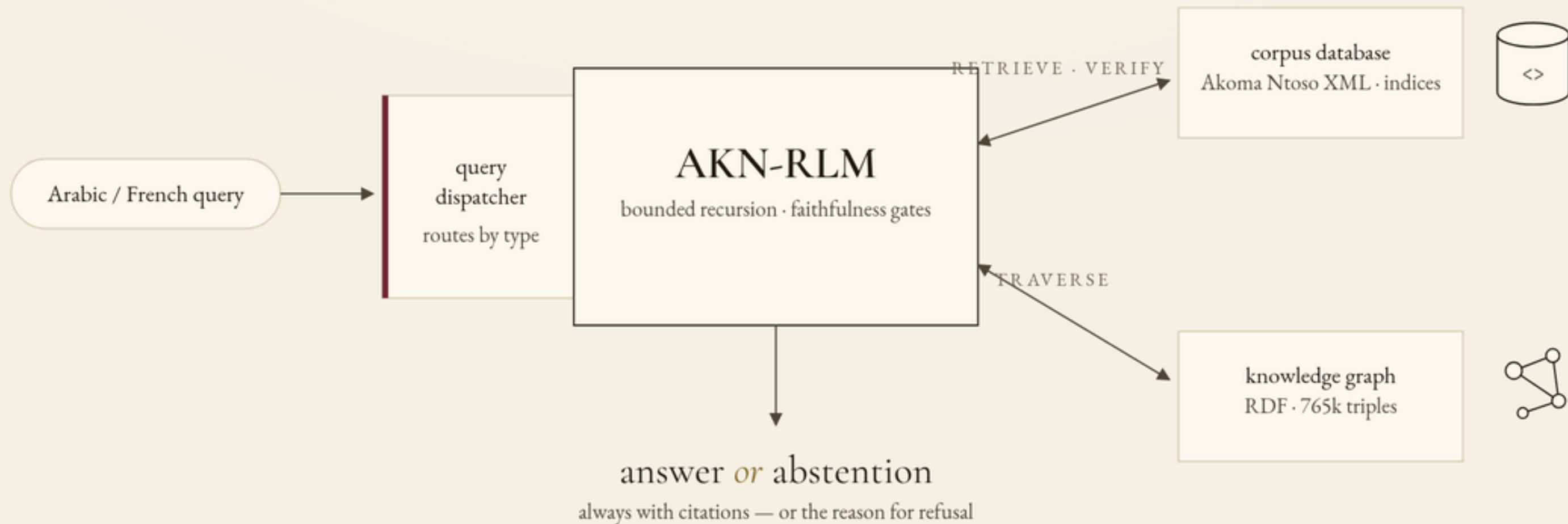
→ AKN-RLM

AKN-RLM

the architecture, layer by layer

THE SYSTEM IN CONTEXT

AKN-RLM in *context*



Nothing reaches the user that did not originate, resolve, and verify inside the system.

CONTRIBUTION I — CORPUS & KNOWLEDGE GRAPH

From the Official Journal to *addressable law*



171 files · 189.46 MB · 1963–2025 — 35 PDF · 45 text · 45 AKN XML · 46 RDF

```
<!-- Civil Code — Ordinance 75-58 -->
<act name="code_civil">
  <FRBRWork value="/akn/dz/act/1975-09-26/75-58"/>
  <article eId="art_124">
    <num>124 المادة</num>
    <content>كل فعل أيا كان</content>
    <ref href="#art_127"/>
  </article></act>
```

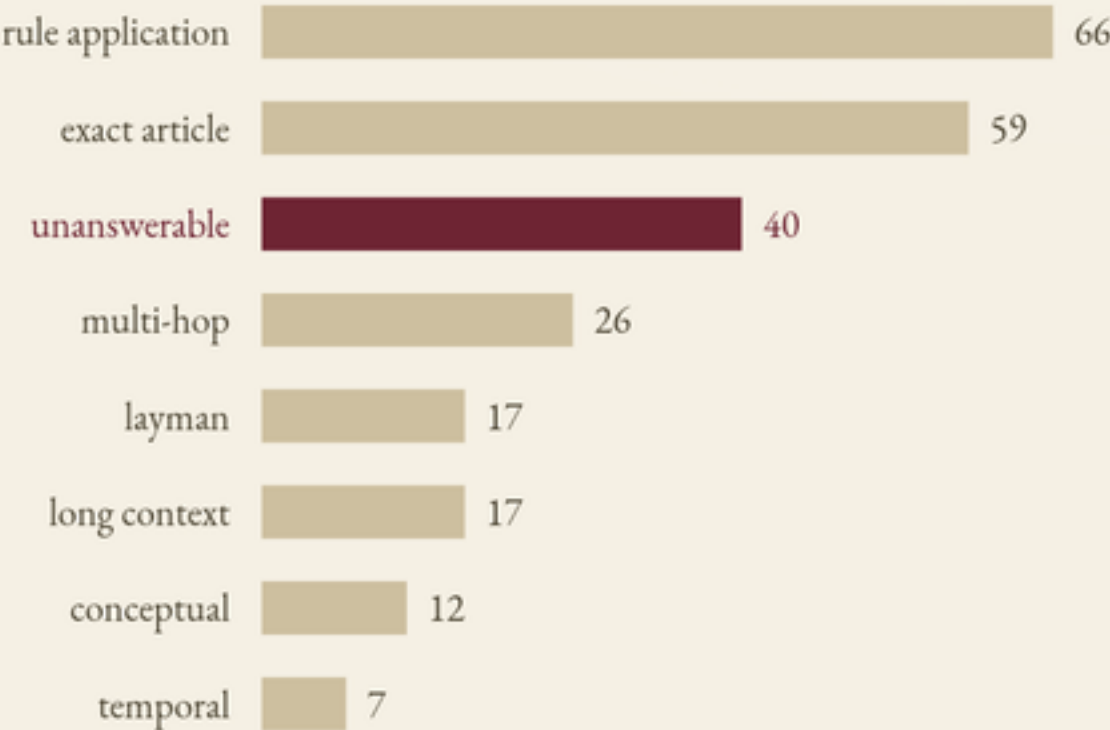
EXPERT VALIDATION

Validated in multiple manual passes by two law students and their supervising senior legal expert — the structure is derived from the sources, never invented.

CONTRIBUTION II — THE EVALUATION LAYER

AlgerianLegalBench

244 questions · 23 legal categories · 8 query types
232 ARABIC / 12 FRENCH



ANNOTATION PROTOCOL — THE SAME EXPERTS WHO
VALIDATED THE CORPUS

- i a law student drafts — question · ground-truth answer · reasoning chain
- ii a second annotator re-derives independently — from the legal texts alone
- iii the senior legal expert resolves disagreements

inter-annotator agreement $\kappa = 0.829$ · 177 answerable / 67 unanswerable · 40 infection traps

GOLD SIGNALS ON EVERY ITEM

expected documents

expected articles

answerability

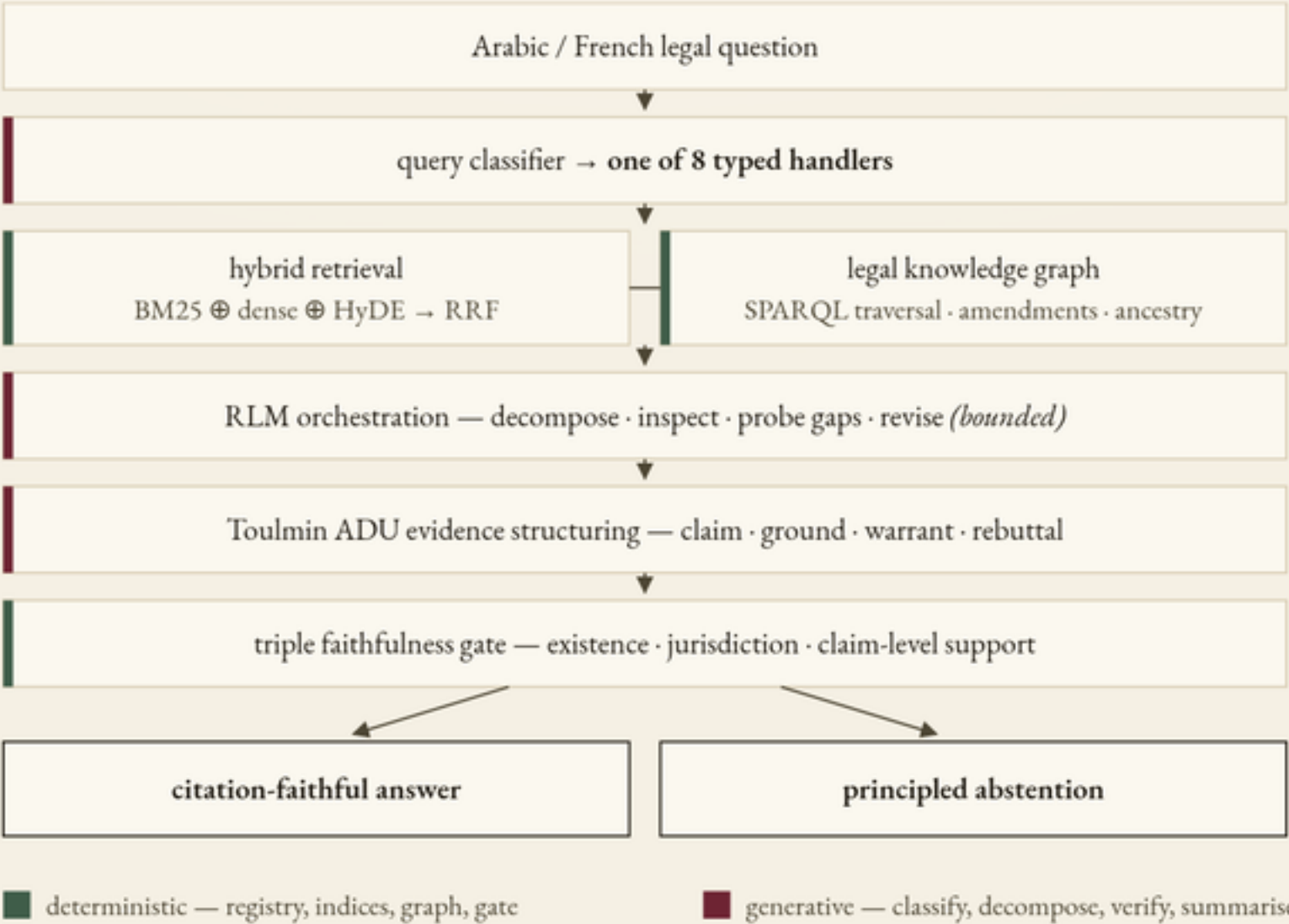
reference answer

4-step reasoning chain

A benchmark built to falsify our own system — retrieval, citation, faithfulness and abstention are all measurable.

CONTRIBUTION III — THE ARCHITECTURE

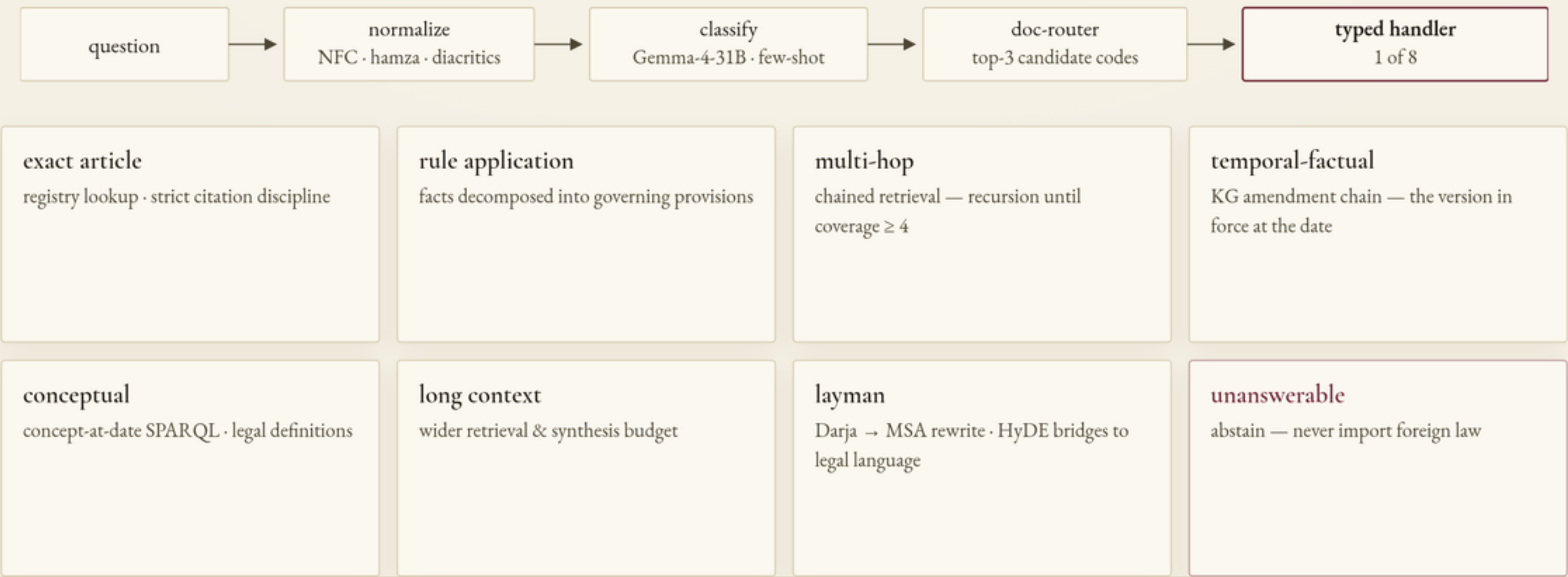
AKN-RLM at a glance



Not a chatbot, not plain RAG — every stage is typed, bounded, and verified before delivery.

THE QUERY DISPATCHER

One question, *eight behaviours*

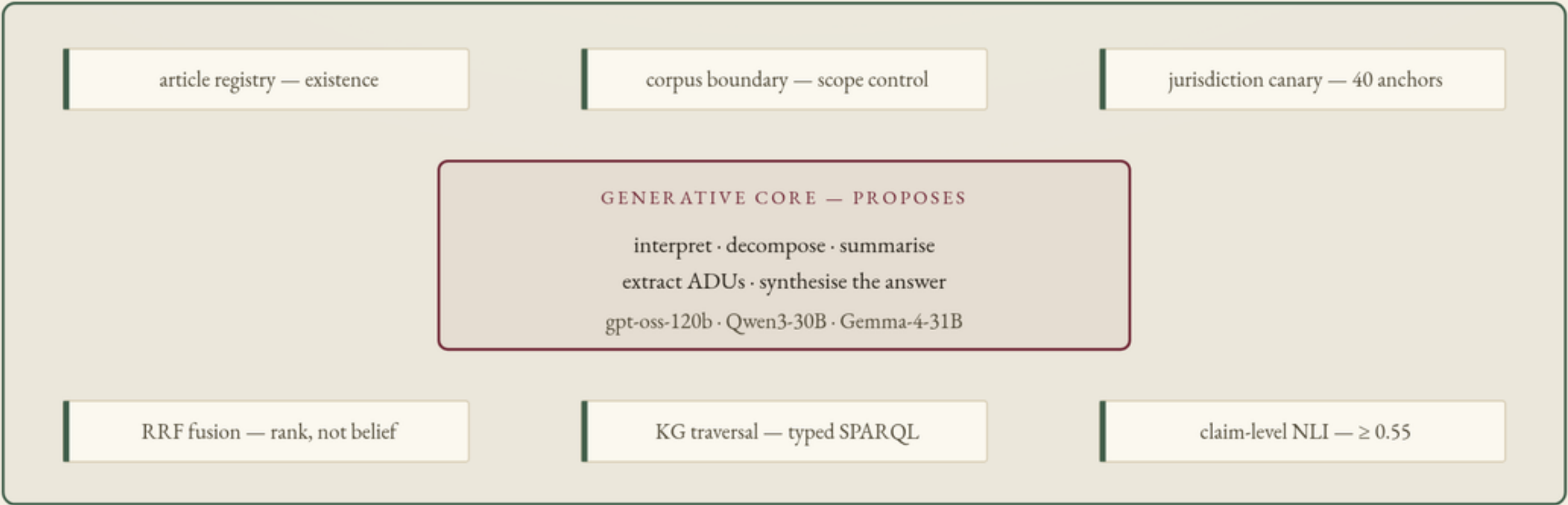


A temporal question and a lay question fail differently — so they are not handled identically.

DESIGN PRINCIPLE

Generative inside, *deterministic outside*

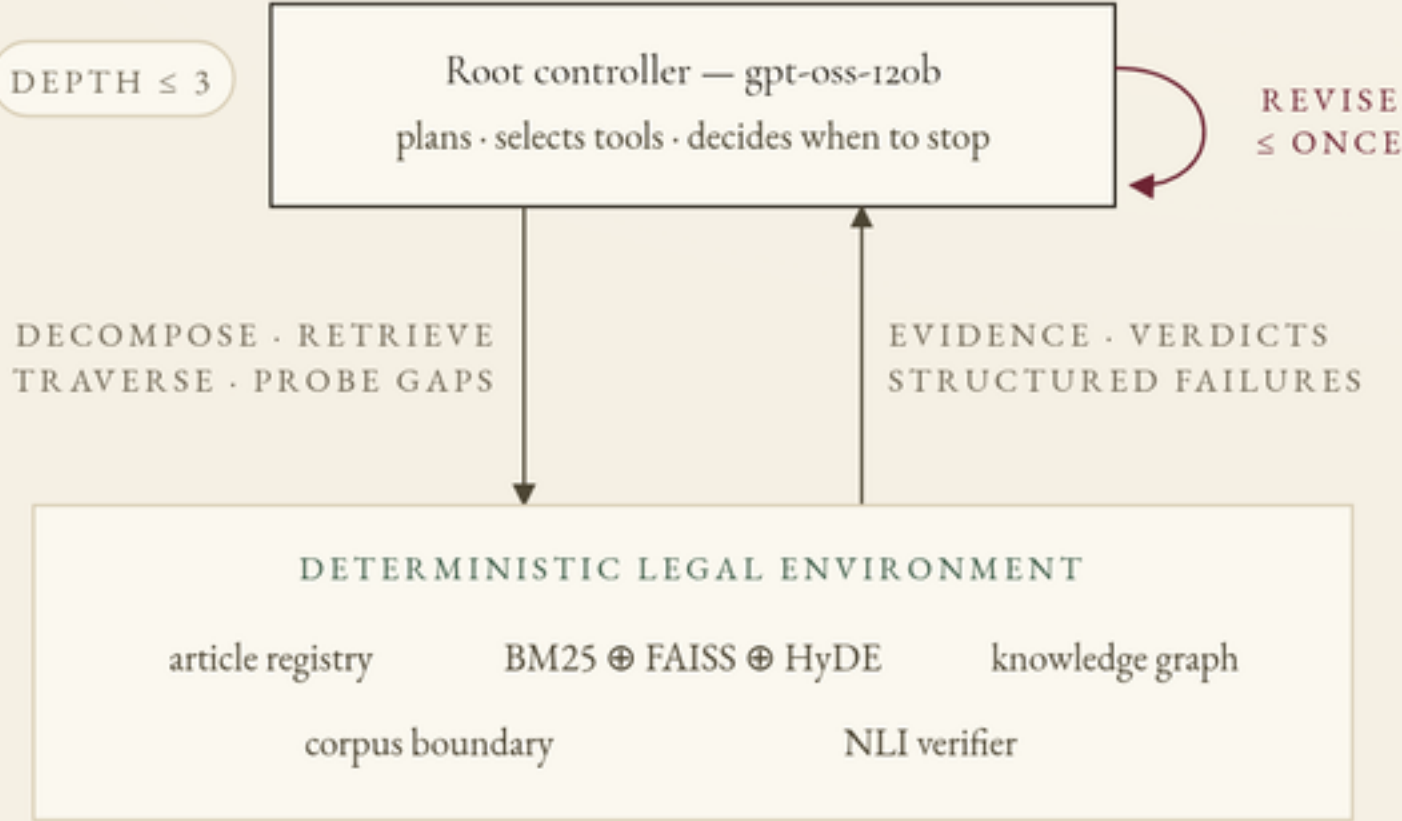
DETERMINISTIC SHELL — DEFINES WHAT MAY BE ASSERTED



The LLM proposes; the infrastructure disposes — language models never define legal authority.

ORCHESTRATION

Bounded recursion — the right to *look again*



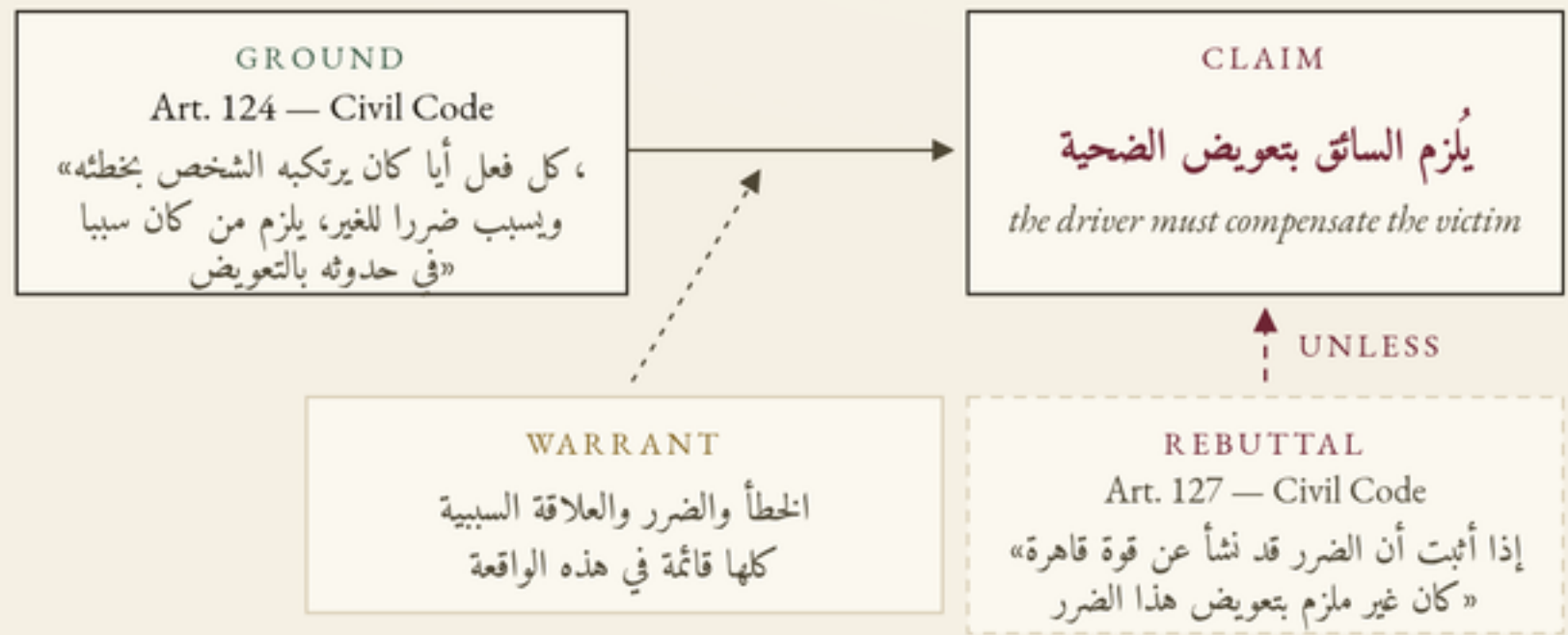
exercised hardest on multi-hop and temporal questions

- i* **Gap-probe** — after each pass, the root model names the element still unsupported — and retrieves for it specifically
- ii* **Coverage floor** — multi-hop & rule-application synthesise only once ≥ 4 supporting articles are held
- iii* **Bounded workers** — decomposition, verification & summaries delegated to Qwen3-30B, structured JSON calls
- iv* **Audited** — every step, retry and verdict lands in the trajectory — telemetry is part of the result

Recursion buys the right to look again — at most three times, and only through the deterministic environment.

EVIDENCE STRUCTURING

Toulmin-style ADUs — evidence that can be *audited*



every unit carries its source article — nothing floats free

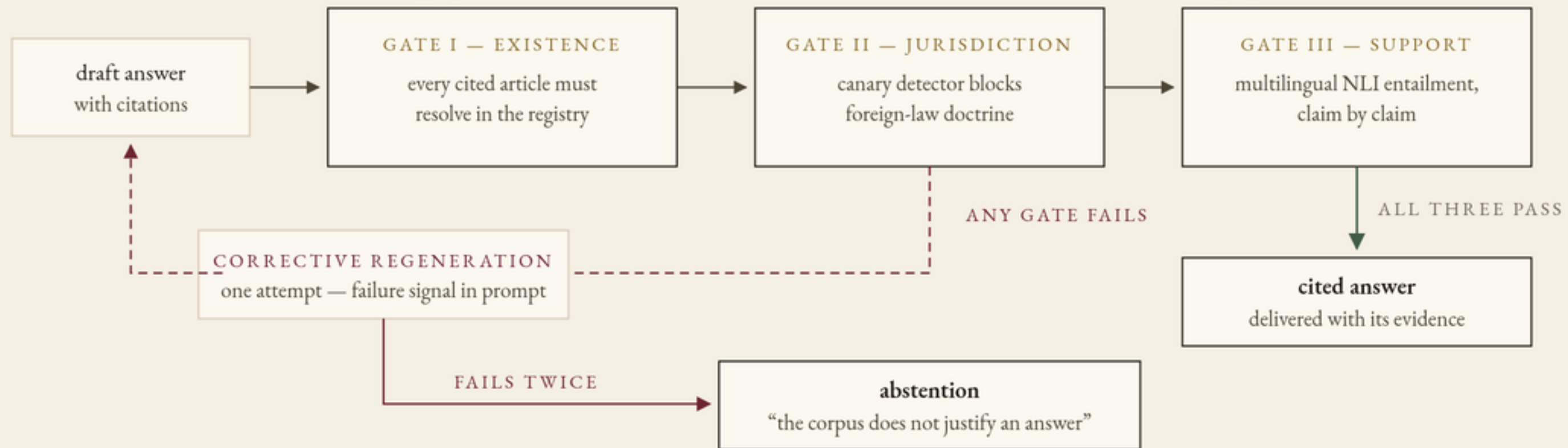
- i* **Pervasive** — Gemma-4-31B extracts ADUs from every retrieved article, not just the final answer
- ii* **Roles survive generation** — rule, condition, exception cannot collapse into fluent prose
- iii* **Feeds gate iii** — each claim is NLI-checked against its own grounds (≥ 0.55)
- iv* **Makes faithfulness measurable** — mirrors the benchmark’s 4-step reasoning chain — AMF 0.471

a lightweight structuring layer — not a full argument-mining system; no annotated Algerian argument corpus exists.

Flat text cannot be audited — structured evidence is what makes faithfulness measurable.

VERIFICATION

The triple faithfulness gate



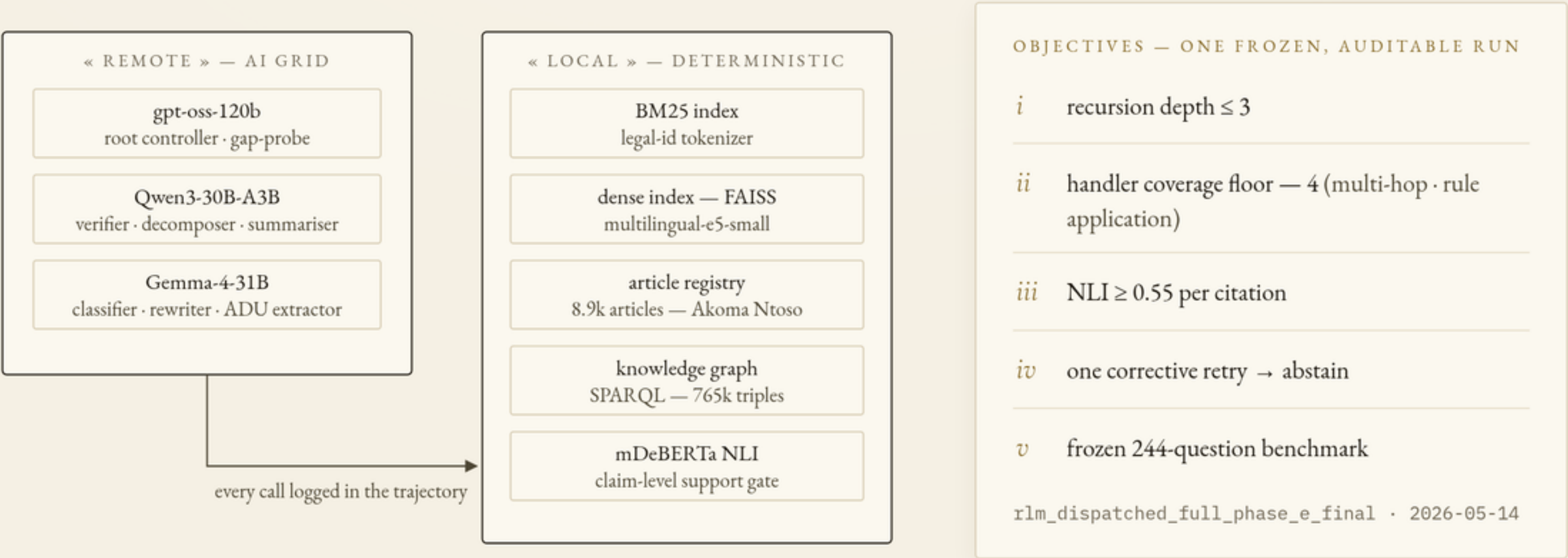
Abstention is a first-class legal answer — refusing is safer than inventing.

Evaluation & Results

deployment · metrics · what the locked run shows

DEPLOYMENT & OBJECTIVES

The locked configuration



Heavy generation runs remotely — everything that defines legal validity runs locally, deterministically.

EVALUATION SETUP & METRICS

Measured on every axis it could fail

TIER 1 — DIRECT LLMS			TIER 2 — RETRIEVAL	TIER 3 — MINIMAL RAG	PHASES & ABLATIONS
seven models, no corpus — <i>are LLMs alone enough?</i>			BM25 · dense · hybrid · rerank · KG — <i>is retrieval enough?</i>	gpt-oss-120b + top-5 — <i>is retrieve-then-generate enough?</i>	layer by layer — <i>which components actually pay?</i>
FAMILY	METRICS		QUESTION IT ANSWERS		
Retrieval	nDCG, MRR, recall@k — article level		did the evidence even arrive?		
Citation	precision / recall / F1 at article granularity		is the right provision cited?		
Generation	BLEU, ROUGE, BERTScore		is the answer text adequate?		
Faithfulness	hallucinated-citation rate, jurisdictional-infection rate, NLI support		is every claim covered by real law?		
Abstention	abstention precision / recall / F1		does it refuse when it should?		
Telemetry	RLM steps, retries, gate verdicts		what did the controller actually do?		

Article-level Citation F1 is the headline — document-level metrics hide “right law, wrong article.”

Main results (1/2)

0.305

ARTICLE CITATION F1

0.546 at document level

0.000

HALLUCINATED-CITATION RATE

every citation resolves in the registry

0.000

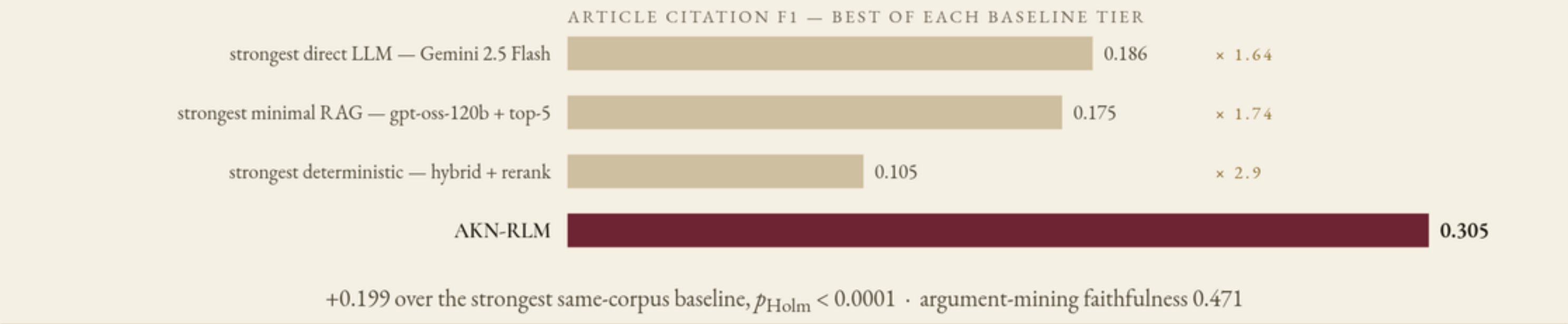
JURISDICTIONAL-INFECTION RATE

no foreign-law doctrine imported

0.703

ABSTENTION F1

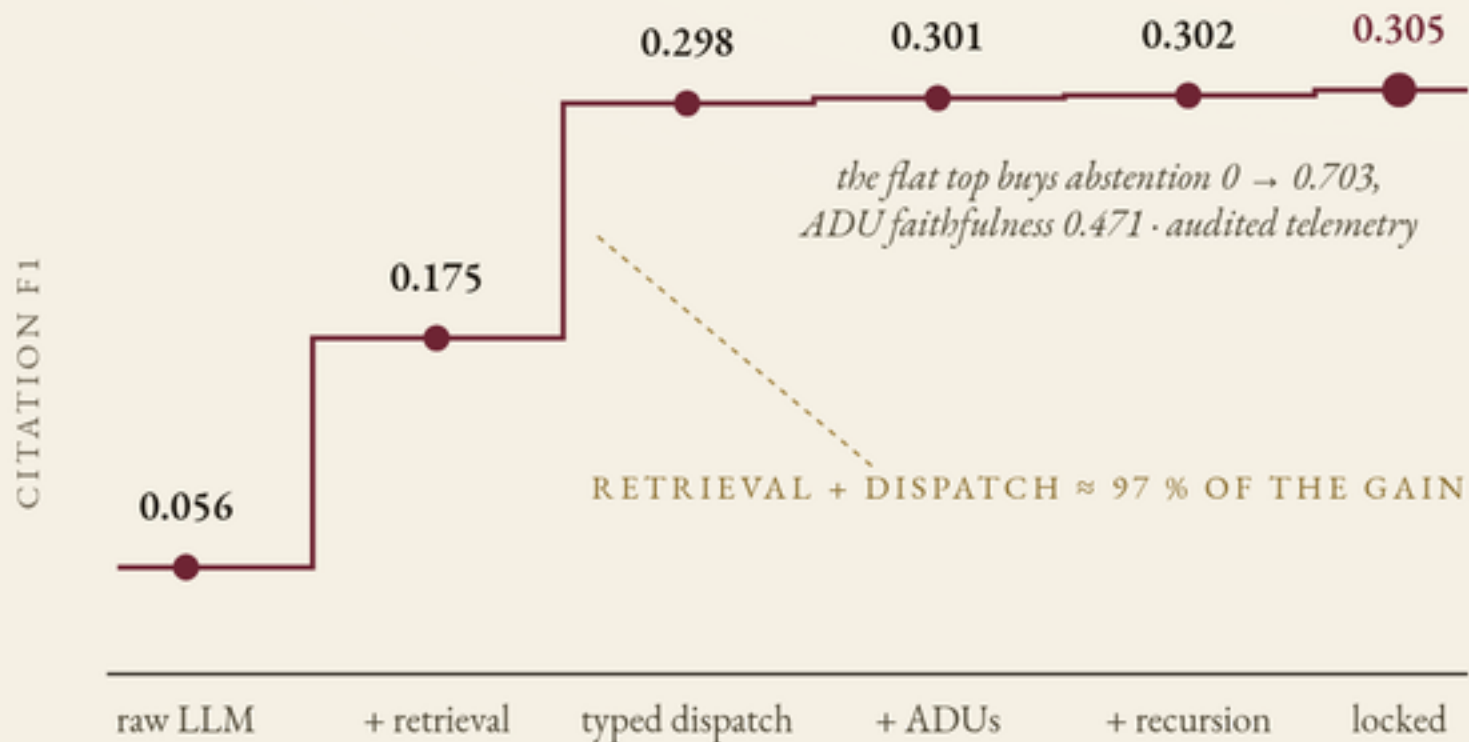
the only compared system that can refuse



Citation quality and zero fabrication, in the same system — that combination is the contribution.

READING THE RESULTS

Interpretation (2/2)



- i **The gain is architectural** — same corpus, registry and indices as every tier-2 baseline
- ii **The ceiling is retrieval** — gold article in the top-10 on $< 30\%$ of queries; infrastructural, not generative
- iii **French is the hard slice** — an Arabic-primary system today
- iv **Strictness has a price** — the gate over-rejects layman queries; wider evidence pools dilute citation discipline
- v **Refusal works** — 38 / 40 infection traps correctly refused, where every baseline invents an answer

Each phase is an ablation in disguise — every layer pays for itself, though not always in F1.

Conclusion

what we claim — and what we do not

THE RESEARCH QUESTION — ANSWERED

Reliability tracks *legal structure*, not parametric scale

CONTRIBUTION	WHAT IT ESTABLISHES
Structured Algerian corpus	45 legal texts, 1963–2025, as Akoma Ntoso — addressable articles · stable eIds · registry
Legal knowledge graph	765k RDF triples over 8.9k articles — references, amendments, hierarchy, versions
AlgerianLegalBench	244 questions · 23 categories · 8 query types — gold articles, reasoning chains, 67 unanswerable
AKN-RLM architecture	typed handlers · hybrid retrieval · KG traversal · bounded recursion · ADUs · triple gate
Empirical evidence	Citation F1 0.305, above every tier — 0.000 fabrication · 0.000 infection · 0.703 abstention

The framework is contextual *and* methodological: built for Algeria, designed as a general recipe for **low-resource, citation-faithful legal AI**. No claim of legal advice; a controlled but realistic subset of Algerian positive law.

Not a chatbot that sounds legal — a system whose answers stay anchored in verifiable Algerian legal sources.

Future work — *where to push next*

- i* **Enlarge the corpus** — from 45 foundational texts toward the breadth of Algerian positive law — the pipeline and registry already scale

- ii* **Deepen the knowledge graph** — let graph relations drive candidate selection directly: multi-hop, temporal, amendment-sensitive questions

- iii* **Raise the retrieval ceiling** — article Recall@10 < 0.30 is the binding constraint today; corpus-tuned embeddings, stronger article ranking

- iv* **Go fully multilingual** — first-class French and lay-Darja coverage, through parallel and French-facing legal material

- v* **Put legal experts in the loop** — systematic expert review of system outputs, beyond automated proxies — with the experts who built the benchmark

The limits are measured, not suspected — which is exactly what turns them into a work programme.



Thank you · Merci · شكراً

Moving on to the business Pitch!

Ibrahim El Khalil Attia · Maab Chaoui

AKN-RLM · ENSIA · JUNE 2026